Tashkent Transit & Walkability: Who Can Reach the Metro in Ten Minutes?

A pedestrian-network model of walking access to metro, buses and everyday destinations across 12 districts

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STATS 401 — Data Visualization · Final Project

Explore the live interactive site:
tashkent-transit-walkability.saidkamolbahodirov.workers.dev

Motivation & question

A metro line on a map does not mean residents can *reach* it on foot. Rail corridors, canals and walled blocks make two points the same crow-flies distance a very different walk apart.

How much of Tashkent's analysed population is within a modelled ten-minute walk of metro access, and where does the bus network fill the gap?

Five questions: (1) citywide metro share; (2) district variation; (3) bus-only dependence; (4) density vs access; (5) bazaar–transit overlap.

The ten-minute model

- **10 minutes at 4.8 km/h = 800 m** of distance along the OpenStreetMap pedestrian network — not a straight-line circle.
- Each population cell is routed to the nearest metro access point and bus stop; metro routing is **entrance-aware** (154 access points: 144 entrances, 10 station-point fallbacks).
- Three **mutually exclusive** classes that sum to 100%: metro / bus-only / underserved.

Data & method

- **OpenStreetMap** (ODbL): pedestrian network, stations, entrances, bus stops, bazaars, boundaries.
- **SIAT 2026-Q2**: official district population totals.
- **WorldPop R2025A**: distributes each district's official total within its boundary only.
- **12 matched districts; 3,212,200 residents.** Yangi Toshkent is excluded — no official total to calibrate against.
- Python geospatial analysis; static D3 + MapLibre dashboard.

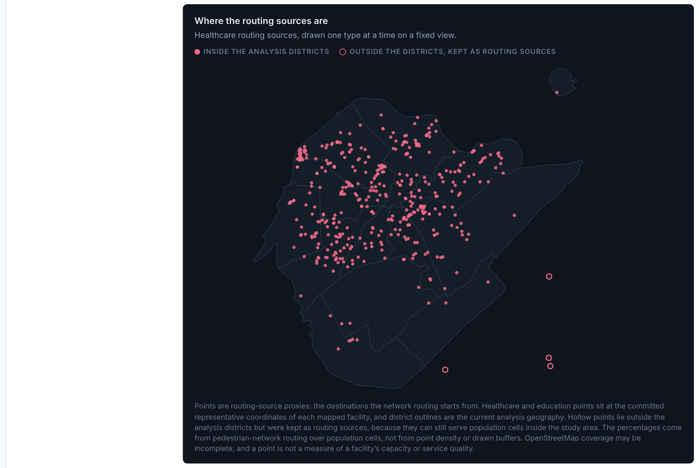
What the model does *not* claim

- Physical walking proximity, **not** observed trips, frequency, fares, transfers or service quality.
- Cell-to-network connectors are straight lines that may cross unmapped barriers.
- Headline moves with walking speed: **10.0% at 4.0 km/h → 17.3% at 5.6 km/h**.
- Mapped bazaars/facilities are an OSM lower bound, not a census of capacity or quality.

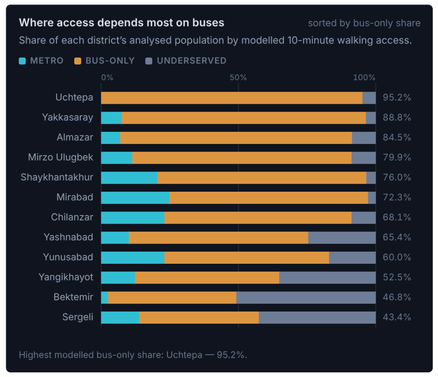
13.6%
Metro access
435,690 residents

72.0%
Bus-only
2,314,113 residents

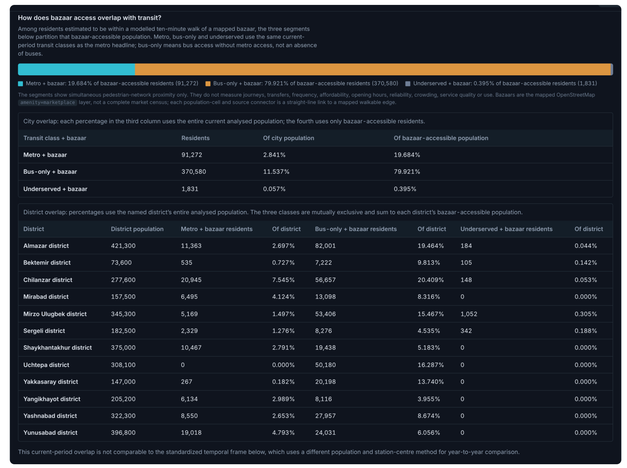
14.4%
Underserved
462,397 residents



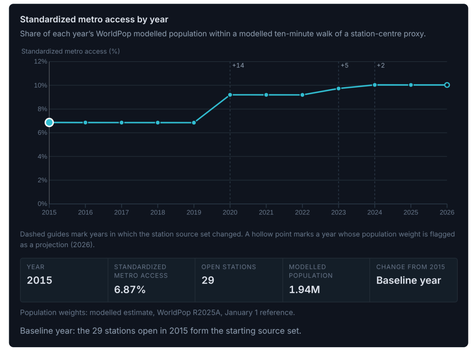
Geographic model. District outlines and mapped routing-source points; access follows the pedestrian network, not a circle. (Live site shows the full MapLibre access map with the modelled service area and density grid.)



Dependence is uneven. Districts sorted by bus-only share — **Uchtepa 95.2%** bus-only; metro access ranges from **Mirabad 24.9%** to **Uchtepa 0.0%** (its nearest cell is only ~19.7 m beyond the 800 m budget).



Bazaar–transit overlap (Q5). Of 463,682 residents near a mapped bazaar: **19.7% metro+bazaar**, **79.9% bus-only+bazaar**, **0.4% underserved+bazaar** — conditional shares of bazaar-accessible residents, kept separate from city-population shares.



Standardized history (separate method). Modelled access rises 6.9% (2015) → 10.0% (2026) using station-centre proxies and annual weights — **not comparable** to the entrance-aware 13.6% headline.

Everyday destinations

Same 800 m model: **46.9%** reach mapped healthcare, **72.5%** education, **14.4%** a bazaar. The committed healthcare-and-education intersection is **43.7%** (residents reaching both, not a sum).

Takeaway

On this model, metro walking access in Tashkent is **narrow and concentrated along the lines** — about one analysed resident in seven. The large majority can walk to a bus stop but not the metro, and that dependence is very unevenly spread across districts. Everything is reproducible from audited outputs, with the uncertainty on the page rather than buried.